SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

CHENNAI – 602105

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – INDUSTRY PROBLEM SOLVING

Course Code:	DSA0502	**Course Title:**	Query Processing for Data Science
CO Assessed:	CO5	**Assessment:**	Tool 2 – INDUSTRY PROBLEM SOLVING
Weightage:	35%	**Total Marks:**	
CO5 Statement: Create meaningful visualizations using Pandas and Matplotlib, including charts, distributions, relationships, and interactive visual representations for effective communication			
Student Name:	U.Ananya	**Reg. No.:**	192424382
Section / Batch:	AI&DS / 24 Batch	**Date of Submission:**	28-05-2026
Faculty In-charge:	K. Veena Devi	**Project Mode:**	Individual Submission

Code of Conduct

I __U.Ananya(192424382______ (Name / Reg No) certifies that this submission is my original work and that I have adhered to all guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature: ___U.Ananya__________

2. E-Commerce Sales Pattern Analysis

Objective

To analyze e-commerce sales data and identify patterns in quantity sold, revenue, discount, and customer ratings. The analysis uses different visualizations to understand sales trends and relationships between variables.

Dataset Columns

- Date
- ProductCategory
- Quantity
- Revenue
- Discount
- CustomerRating

Python Code

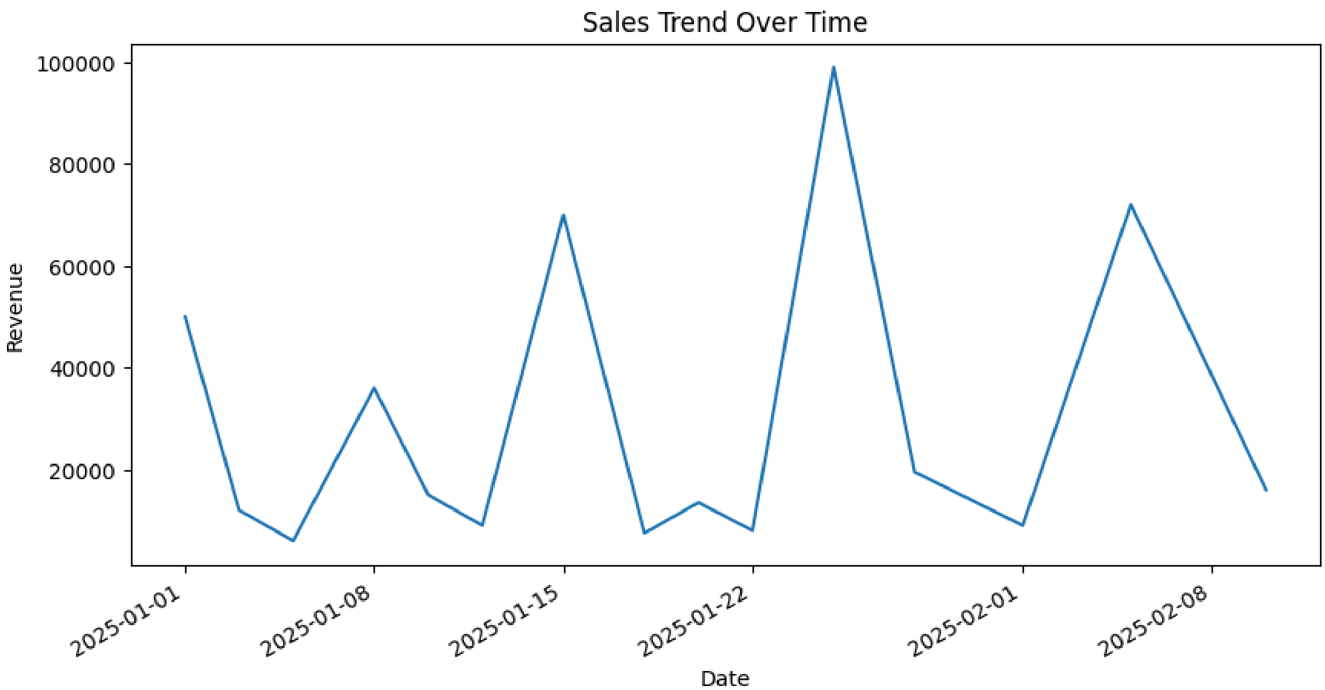
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
# Load dataset
df = pd.read_csv("ecommerce_sales.csv")
print(df.head())
print(df.describe())
# Convert date
df["Date"] = pd.to_datetime(df["Date"])
# 1. Sales trend over time
daily_sales = df.groupby("Date")["Revenue"].sum()
plt.figure(figsize=(10,5))
daily_sales.plot()
plt.title("Sales Trend Over Time")
plt.xlabel("Date")
plt.ylabel("Revenue")
```

```python
plt.show()
# 2. Revenue by category
category_sales = df.groupby("ProductCategory")["Revenue"].sum()
category_sales.plot(kind="bar", figsize=(8,5))
plt.title("Revenue by Product Category")
plt.xlabel("Category")
plt.ylabel("Revenue")
plt.show()
# 3. Quantity distribution
sns.histplot(df["Quantity"], kde=True)
plt.title("Quantity Sold Distribution")
plt.show()
# 4. Discount frequency
sns.countplot(x="Discount", data=df)
plt.title("Discount Frequency")
plt.xticks(rotation=45)
plt.show()
# 5. Discount vs Revenue
sns.scatterplot(
    x="Discount",
    y="Revenue",
    data=df
)
plt.title("Discount vs Revenue")
plt.show()
# 6. Quantity vs Revenue
sns.scatterplot(
    x="Quantity",
    y="Revenue",
    data=df
)
plt.title("Quantity vs Revenue")
plt.show()
```

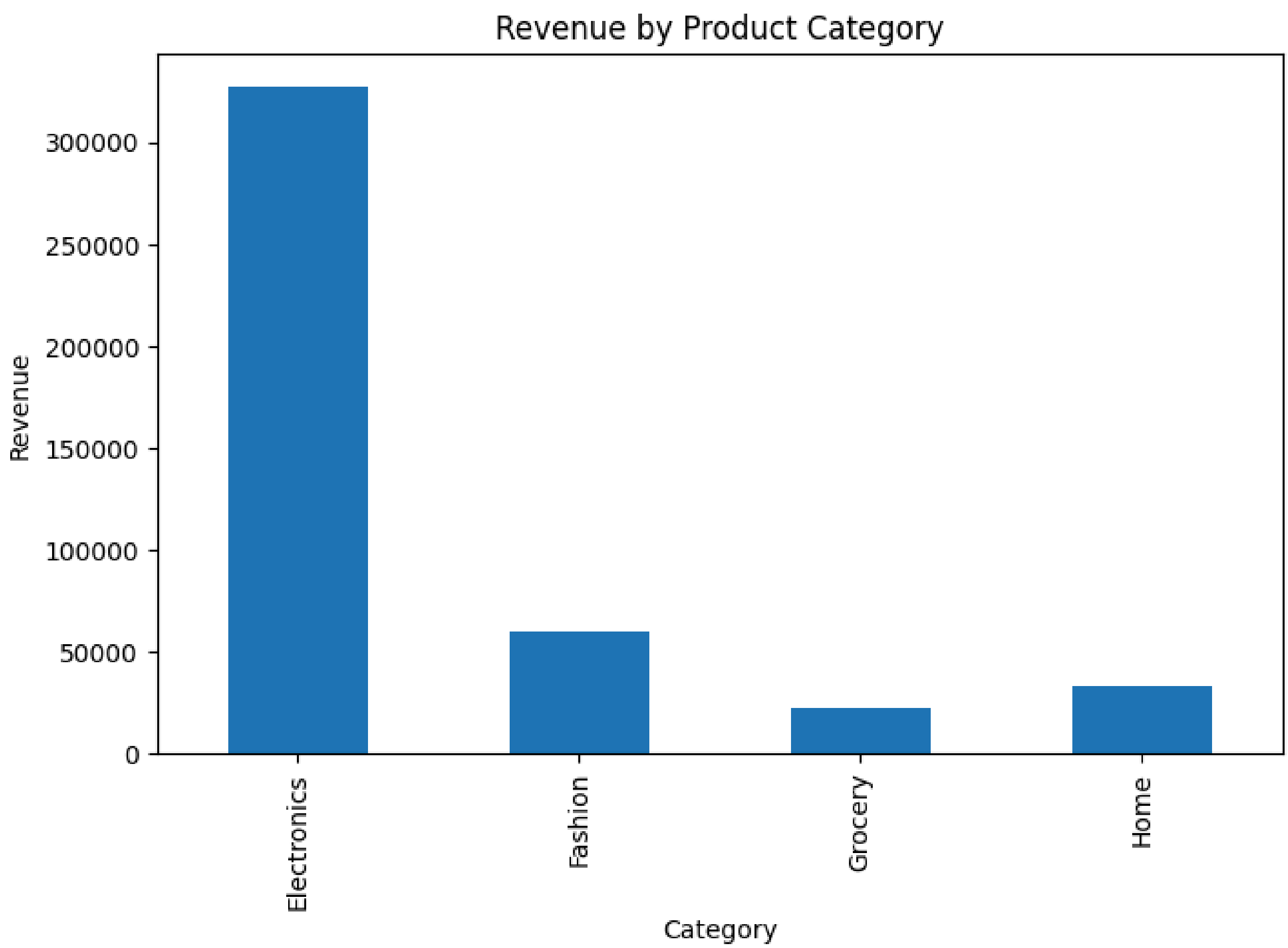
```
# 7. Customer Rating distribution
sns.histplot(df["CustomerRating"], kde=True)
plt.title("Customer Rating Distribution")
plt.show()
# 8. Scatter Matrix
pd.plotting.scatter_matrix(
    df[
        ["Quantity", "Revenue",
         "Discount", "CustomerRating"]
    ],
    figsize=(10,8)
)
plt.suptitle("E-Commerce Sales Scatter Matrix")
plt.show()
print("\nCorrelation:")
print(
    df[
        ["Quantity", "Revenue",
         "Discount", "CustomerRating"]
    ].corr()
)
```

Output:

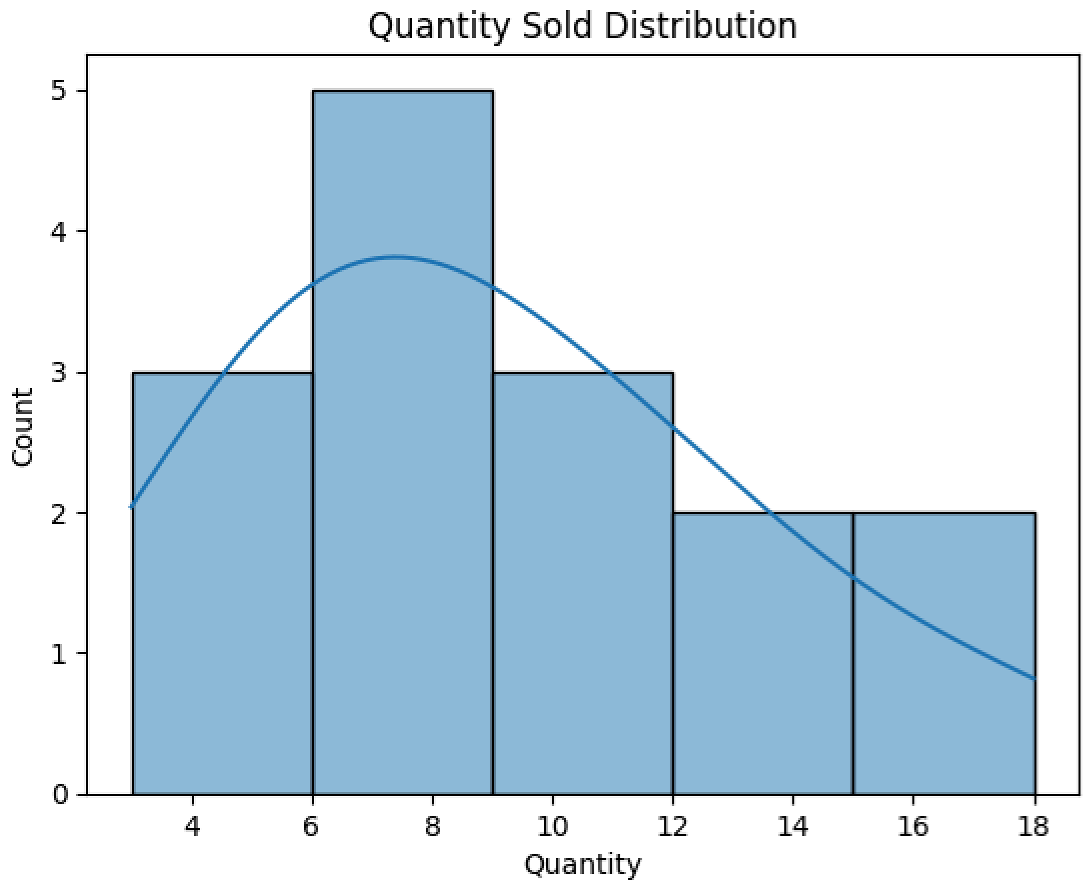
```
         Date ProductCategory  Quantity  Revenue  Discount  CustomerRating
0  2025-01-01     Electronics         5    50000        10             4.5
1  2025-01-03         Fashion         8    12000        15             4.2
2  2025-01-05         Grocery        12     6000         5             4.0
3  2025-01-08     Electronics         3    36000        20             4.7
4  2025-01-10         Fashion        10    15000        10             4.1
        Quantity     Revenue   Discount  CustomerRating
count  15.000000     15.0000  15.000000       15.000000
mean    9.000000  29500.0000  11.666667        4.300000
std     4.208834  29438.7985   5.563486        0.338062
min     3.000000   6000.0000   5.000000        3.800000
25%     6.000000   9000.0000   7.500000        4.050000
50%     8.000000  15000.0000  10.000000        4.200000
75%    11.500000  43000.0000  15.000000        4.550000
max    18.000000  99000.0000  20.000000        4.900000
```

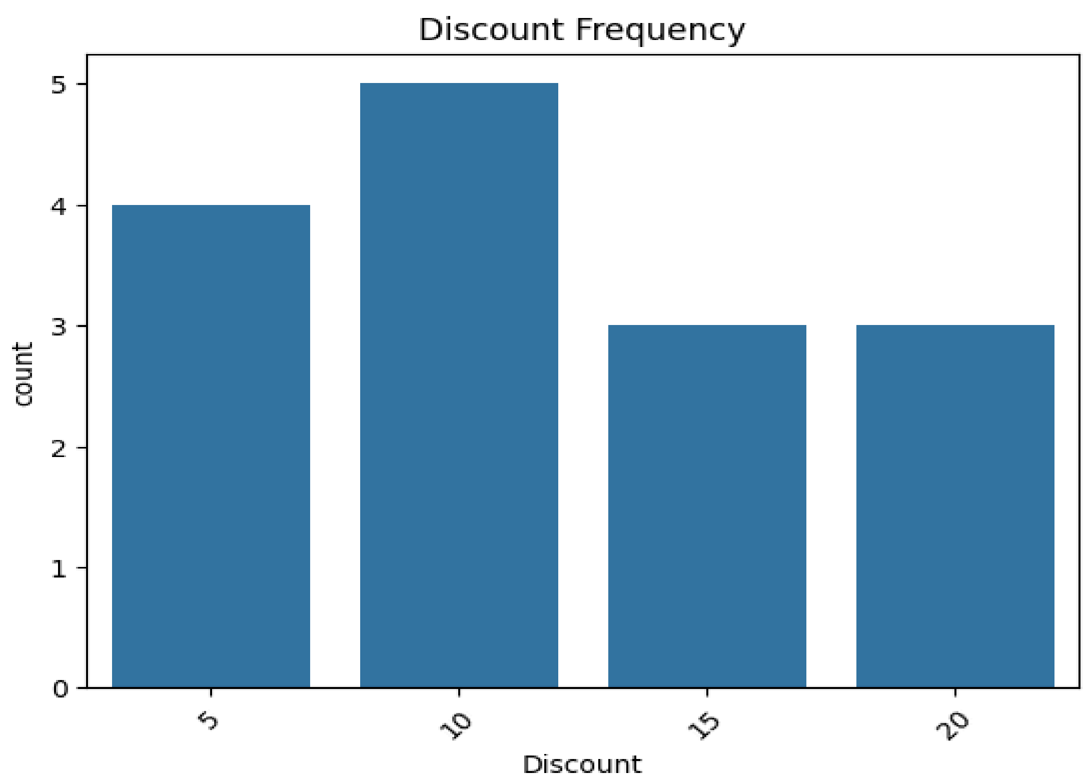


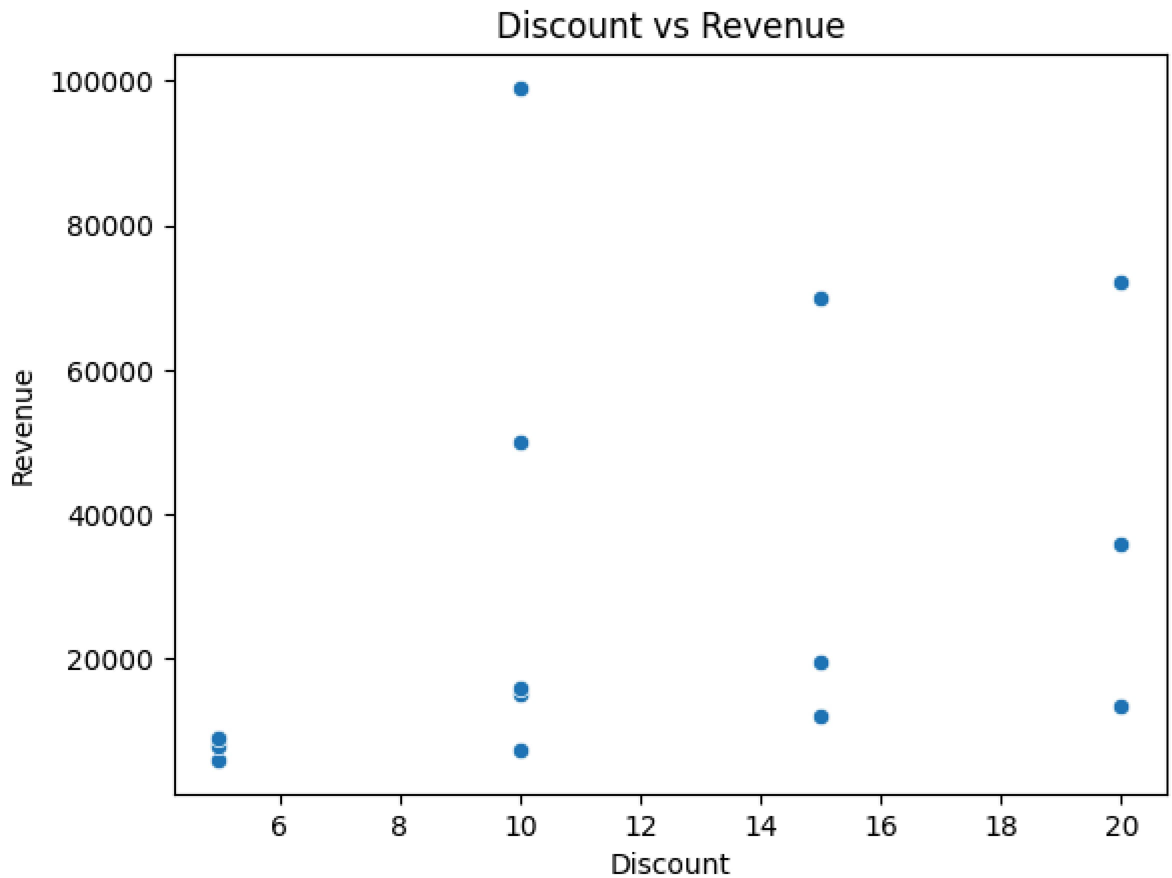
Sales Trend Over Time
Revenue
100000
80000
60000
40000
20000
2025-01-01
2025-01-08
2025-01-15
2025-01-22
2025-02-01
2025-02-08
Date

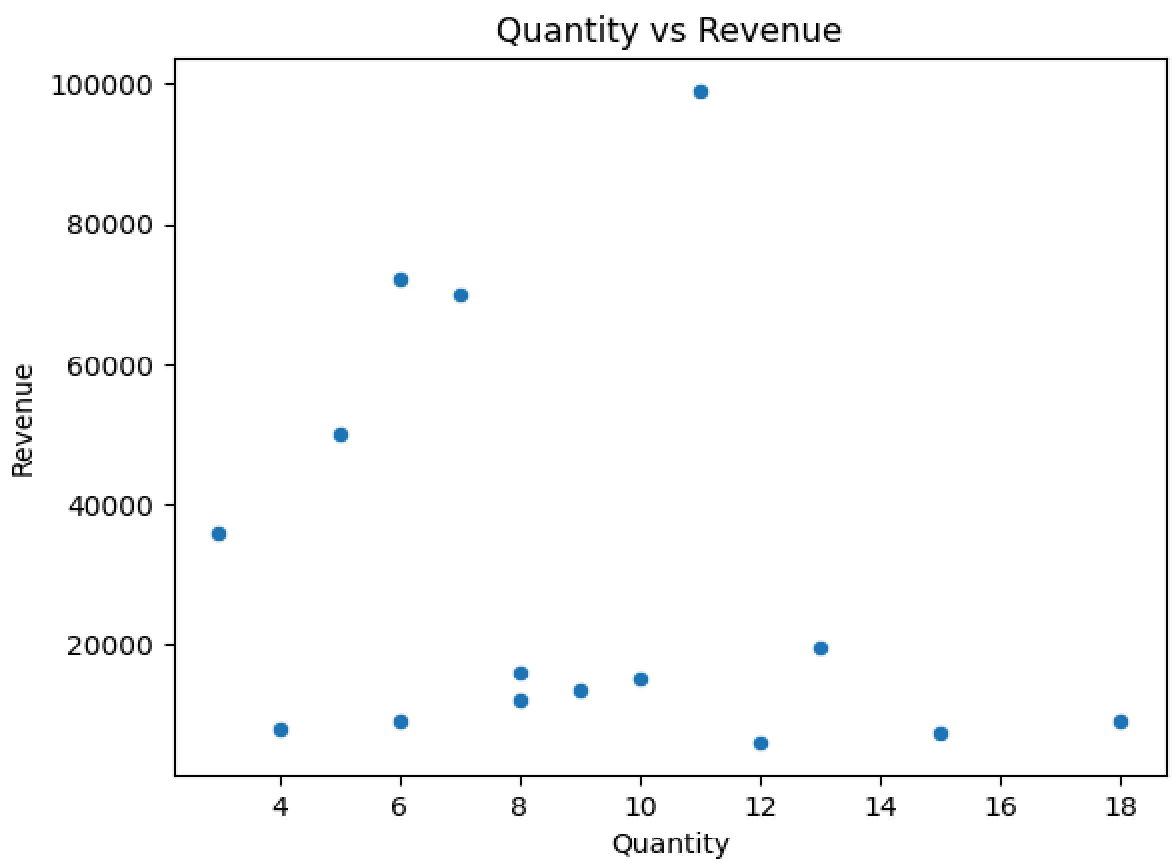


Revenue by Product Category
Revenue
300000
250000
200000
150000
100000
50000
0
Electronics
Fashion
Grocery
Home
Category

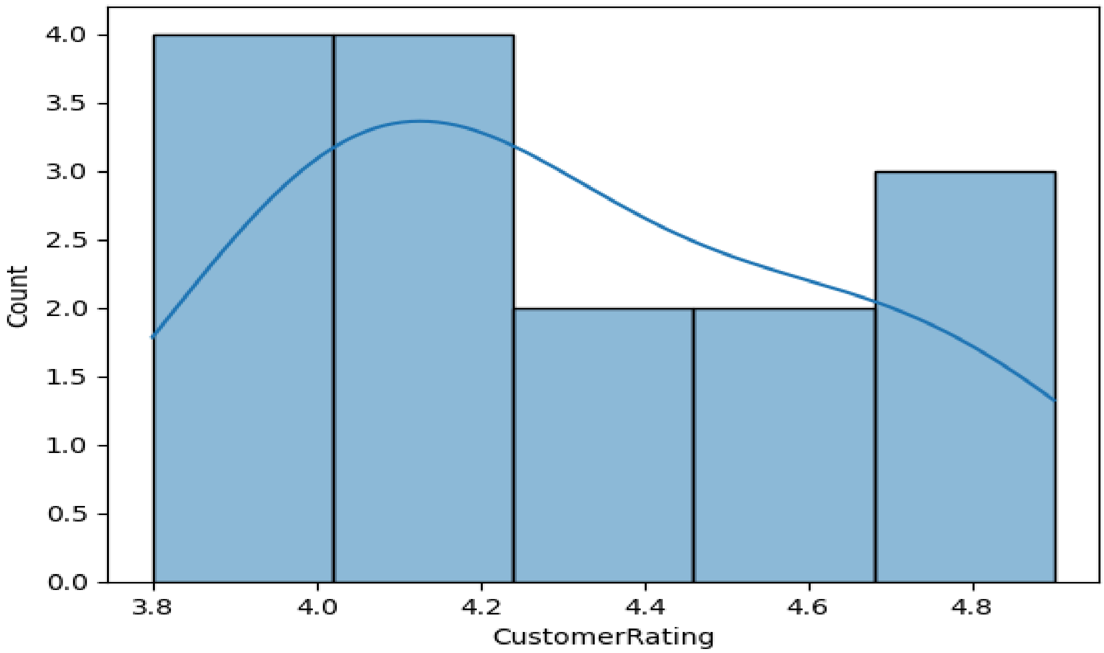

Quantity Sold Distribution
Count
Quantity
0
1
2
3
4
5
4
6
8
10
12
14
16
18


Discount Frequency
count
Discount
0
1
2
3
4
5
5
10
15
20

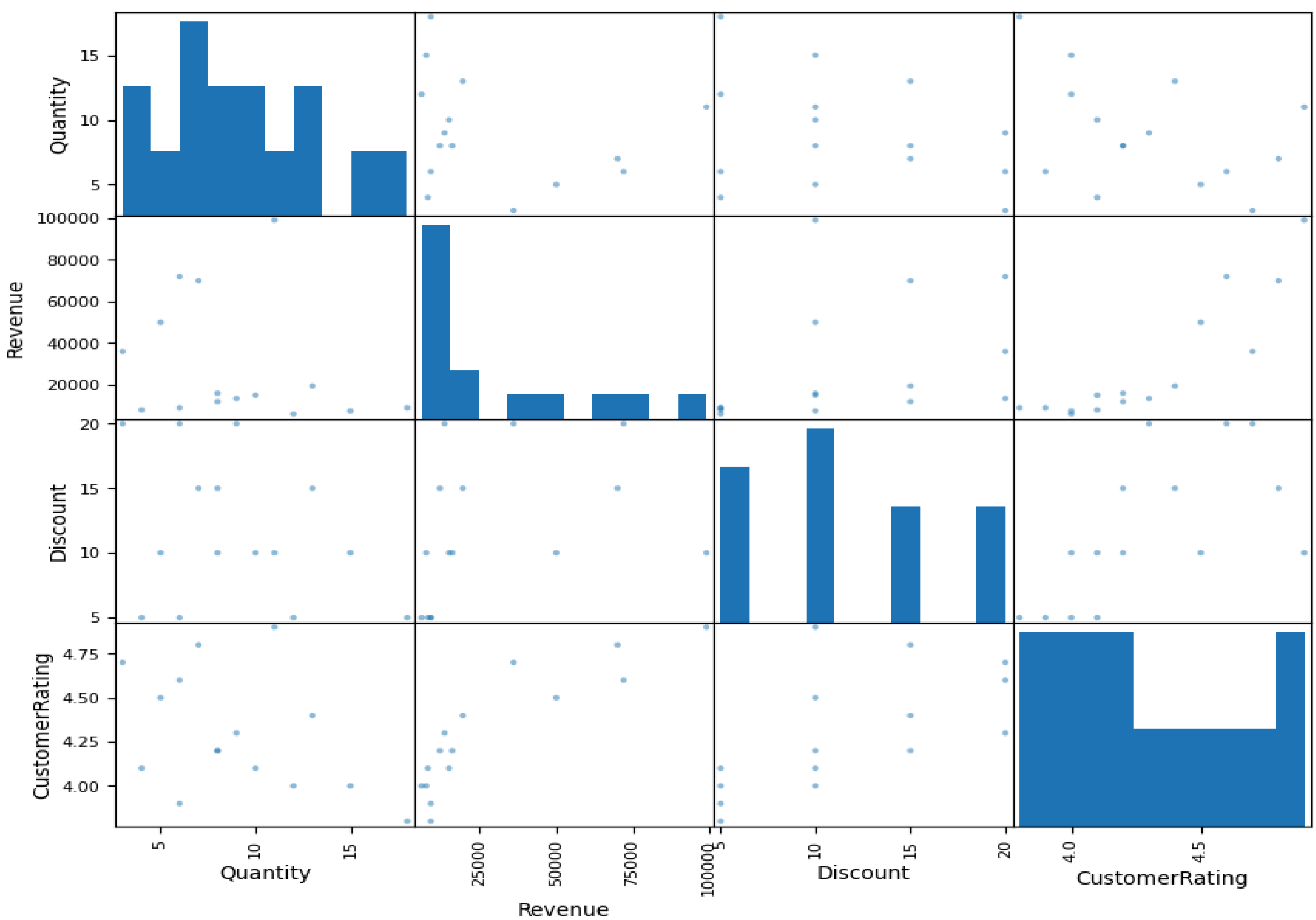

Discount vs Revenue
Revenue
100000
80000
60000
40000
20000
6
8
10
12
14
16
18
20
Discount


Quantity vs Revenue
Revenue
100000
80000
60000
40000
20000
4
6
8
10
12
14
16
18
Quantity

Customer Rating Distribution



E-Commerce Sales Scatter Matrix



```
Correlation:
                Quantity   Revenue  Discount  CustomerRating
Quantity        1.000000 -0.249043 -0.320297       -0.426710
Revenue        -0.249043  1.000000  0.359798        0.881362
Discount       -0.320297  0.359798  1.000000        0.626632
CustomerRating -0.426710  0.881362  0.626632        1.000000
```

Expected Analysis

The dashboard will show:

1. Line Chart – Shows how revenue changes over time.
2. Bar Chart – Compares revenue between product categories.
3. Distribution Plot – Shows the distribution of quantity sold.
4. Frequency Plot – Shows how often different discounts occur.
5. Scatter Plot – Shows the relationship between discount and revenue.
6. Scatter Plot – Shows the relationship between quantity and revenue.
7. Rating Distribution – Shows customer rating patterns.
8. Scatter Matrix – Shows relationships among multiple numerical variables.
9. Correlation Analysis – Measures relationships between sales variables.

Conclusion

The E-Commerce Sales Pattern Analysis helps identify sales trends, popular product categories, customer rating patterns, and relationships between discount, quantity, and revenue. These insights can help the company improve pricing, promotions, inventory management, and overall sales performance.